

Math 122: Exam 1's Solution

Problem 1 Find the exact arc length of the curve $y = (2/3)x^{3/2} + 2026$ over the interval from $x = 0$ to $x = 1$ [cite: 1]. * The formula for arc length is $L = \int \sqrt{1 + (dy/dx)^2} dx$. * Calculate the derivative of the given function: $dy/dx = (2/3) * (3/2)x^{1/2} = \sqrt{x}$. * Square the derivative: $(dy/dx)^2 = x$. * Set up the integral with the given bounds: $L = \int_0^1 \sqrt{1 + x} dx$. * Evaluate using the substitution $u = 1+x$ and $du = dx$: $L = [(2/3)(1+x)^{3/2}]$ from 0 to 1 = $(2/3)(2^{3/2} - 1)$. * Exact answer: $(2/3)(2\sqrt{2} - 1)$ **Decimal answer: 1.21895**

Problem 2 Find the limit: $\lim_{x \rightarrow 0} (2^x - 1) / \sin(2x)$ [cite: 2]. * Substituting $x=0$ yields the indeterminate form $0/0$. * Apply L'Hopital's Rule by taking the derivative of the numerator and the denominator: $\lim_{x \rightarrow 0} (2^x * \ln(2)) / (2 * \cos(2x))$. * Evaluate the limit as $x \rightarrow 0$: $(2^0 * \ln(2)) / (2 * \cos(0))$. * Exact answer: $\ln(2) / 2$ **Decimal answer: 0.34657**

Problem 3 Find the limit: $\lim_{x \rightarrow +\infty} x^2 * 2^{(-3x)}$ [cite: 2]. * Rewrite the expression as a fraction: $\lim_{x \rightarrow +\infty} x^2 / 2^{(3x)}$. * This yields the indeterminate form ∞/∞ , so apply L'Hopital's Rule: $\lim_{x \rightarrow +\infty} 2x / (3 * 2^{(3x)} * \ln(2))$. * Apply L'Hopital's Rule a second time for the remaining ∞/∞ form: $\lim_{x \rightarrow +\infty} 2 / (9 * 2^{(3x)} * (\ln(2))^2)$. * As $x \rightarrow \infty$, the denominator grows without bound while the numerator is constant. * **Decimal answer: 0**

Problem 4 Find the limit: $\lim_{x \rightarrow 0^+} (1+3x)^{(1/x)}$ [cite: 3]. * This limit takes the indeterminate form 1^∞ . * Let $y = (1+3x)^{(1/x)}$. Taking the natural logarithm gives $\ln(y) = \ln(1+3x) / x$. * Evaluate the limit of $\ln(y)$ as $x \rightarrow 0^+$, which takes the form $0/0$, using L'Hopital's Rule: $\lim_{x \rightarrow 0^+} 3/(1+3x) / 1 = 3$. * Since $\lim \ln(y) = 3$, exponentiate to find the limit of y . * Exact answer: e^3 **Decimal answer: 20.08554**

Problem 5 Find du for $y = \arcsin(1/x^3)$ [cite: 3]. * Apply the chain rule using the derivative of inverse sine ($1 / \sqrt{1-u^2}$): $dy/dx = (1 / \sqrt{1 - (x^{-3})^2}) * d/dx(x^{-3})$. * Evaluate the inner derivative: $dy/dx = (1 / \sqrt{1 - x^{-6}}) * (-3x^{-4})$. * **Answer: $dy = -3 / (x^4 * \sqrt{1 - x^{-6}}) dx$** (No decimal evaluation applies).

Problem 6 Find the derivative of $y = \arctan(x) * \operatorname{arcsec}(x)$ [cite: 4]. * Apply the product rule: $dy/dx = d/dx(\arctan(x)) * \operatorname{arcsec}(x) + \arctan(x) * d/dx(\operatorname{arcsec}(x))$. * Substitute the standard derivatives. * **Answer: $dy/dx = (1 / (1+x^2)) * \operatorname{arcsec}(x) + \arctan(x) * (1 / (|x|\sqrt{x^2-1}))$** (No decimal evaluation applies).

Problem 7 Evaluate the integral: $\int 1 / (1+16x^2) dx$ [cite: 4]. * Rewrite the denominator: $\int 1 / (1+(4x)^2) dx$. * Use substitution with $u = 4x$, making $dx = du / 4$. * Substitute and integrate: $(1/4) * \int 1 / (1+u^2) du = (1/4)\arctan(u) + C$. * **Answer: $(1/4)\arctan(4x) + C$** (No decimal evaluation applies).

Problem 8 Evaluate the integral: $\int x^2 / \sqrt{1-x^6} dx$ [cite: 5]. * Rewrite the integral: $\int x^2 / \sqrt{1-(x^3)^2} dx$. * Let $u = x^3$, giving $du = 3x^2 dx$, or $x^2 dx = du / 3$. * Substitute and integrate: $(1/3) * \int 1 / \sqrt{1-u^2} du = (1/3)\arcsin(u) + C$. * **Answer: $(1/3)\arcsin(x^3) + C$** (No decimal evaluation applies).

Problem 9 Evaluate the definite integral: $\int_0^1 (\arcsin x)^4 / \sqrt{1-x^2} dx$ [cite: 5]. * Let $u = \arcsin(x)$, which makes $du = 1 / \sqrt{1-x^2} dx$. * Determine new limits: When $x=0$, $u=0$. When $x=1$, $u=\pi/2$. * Evaluate the new integral: $\int_0^{\pi/2} u^4 du = [u^5 / 5]$ from 0 to $\pi/2$. * Exact answer: $(\pi/2)^5 / 5 = \pi^5 / 160$. * **Decimal answer: 1.91262**

Problem 10 Evaluate the integral: $\int e^{(2x)} / (1+e^{(4x)}) dx$ [cite: 6]. * Rewrite to identify the substitution: $\int e^{(2x)} / (1+(e^{(2x)})^2) dx$. * Let $u = e^{(2x)}$, yielding $du = 2e^{(2x)} dx$, or $e^{(2x)} dx = du / 2$. * Substitute and integrate: $(1/2) * \int 1 / (1+u^2) du = (1/2)\arctan(u) + C$. * **Answer: $(1/2)\arctan(e^{(2x)}) + C$** (No decimal evaluation applies).